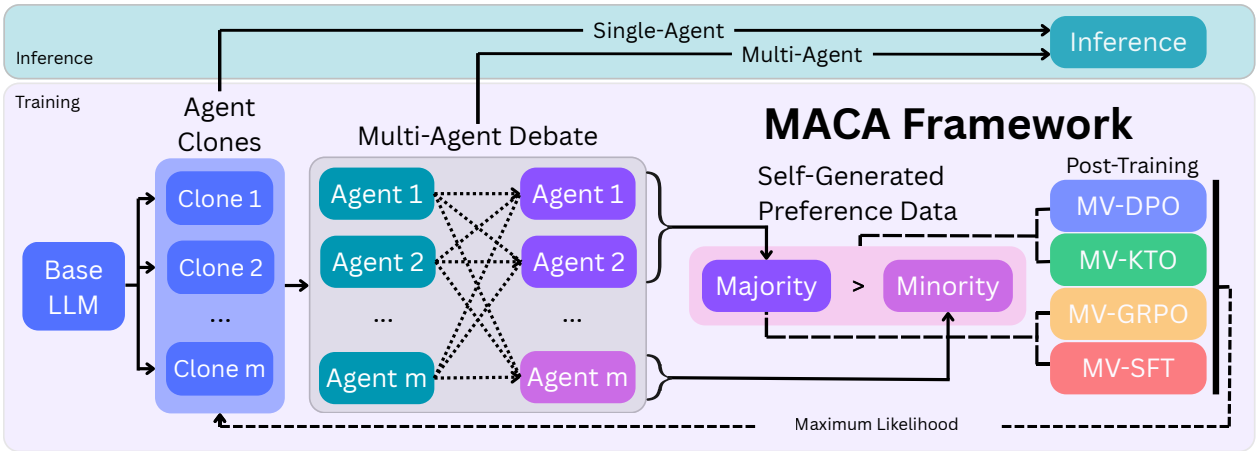


Figure 1 Multi-Agent Consensus Alignment framework: Multiple clones of a base LM engage in debate to generate majority and minority reasoning trajectories through multi-agent debate (collaborative exchange of reasoning). The framework splits responses based on alignment with majority consensus to create preference pairs. MV-GRPO compares online samples against majority signals, while MV-SFT imitates majority traces directly. In contrast, MV-DPO and MV-KTO utilize both positive (majority) and negative (minority) examples to learn to differentiate between these preference pairs. Updated agents can then be used for single-agent or multi-agent inference, or continue iterative training.

sample's answer matches the pre-computed consensus:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi_\theta}\left[\tilde{A}_x(y)\sum_t \log \pi_\theta(y_t|x, y_{<t})\right] + \lambda\,\text{KL}(\pi_\theta\|\pi_{\text{ref}}), \tag{4}$$

where $\tilde{A}_x(y) = r_x(y) - \bar{r}_x$ is the group-normalized advantage. We find that model inconsistency naturally yields both consensus and dissenting trajectories, allowing GRPO's group normalization to contrast majority/minority outcomes within batches.

Majority-Vote DPO (MV-DPO) follows the standard DPO formulation with preference pairs constructed from our pre-generated debate outcomes:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{(y^+,y^-)\in\mathcal{G}^+(x)\times\mathcal{G}^-(x)}\left[\log\sigma\left(\beta\left[\log\frac{\pi_\theta(y^+|x)}{\pi_{\text{ref}}(y^+|x)} - \log\frac{\pi_\theta(y^-|x)}{\pi_{\text{ref}}(y^-|x)}\right]\right)\right]. \tag{5}$$

By contrasting the model's own consensus and dissenting trajectories, DPO's log probability ratios capture differences across entire reasoning chains, not just final answers, allowing each token to contribute to the preference signal.

Majority-Vote KTO (MV-KTO) applies KTO's unpaired formulation with debate-derived labels from our fixed dataset, with class-balancing weights λ_+ and λ_-:

$$\begin{aligned}\mathcal{L}_{\text{KTO}}(\theta) = &-\lambda_+\mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y^+\in\mathcal{G}^+(x)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y^+|x)}{\pi_{\text{ref}}(y^+|x)}\right)\right] \\ &-\lambda_-\mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y^-\in\mathcal{G}^-(x)}\left[\log\sigma\left(-\beta\log\frac{\pi_\theta(y^-|x)}{\pi_{\text{ref}}(y^-|x)}\right)\right].\end{aligned} \tag{6}$$

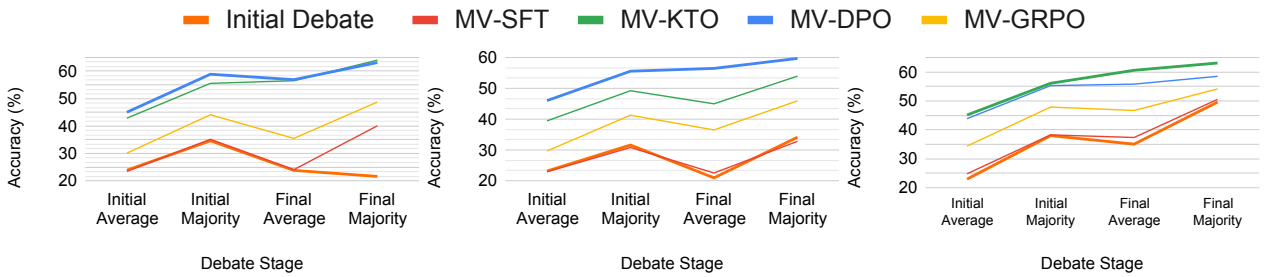
KTO's unpaired structure handles imbalanced outcomes where majority trajectories dominate.

5 Results and Discussion

We evaluate MACA by post-training four instruction-tuned small LMs (Qwen-2B (Yang et al., 2024), Llama-3B (Grattafiori et al., 2024), Phi-4B (Abdin et al., 2024), and Llama-8B (Grattafiori et al., 2024)) on six reasoning benchmarks

Table 1 Post-training multi-agent debate yields consistent accuracy improvements.

Model	Dataset	Pre / SFT (Baseline)		RL (Our Methods)			Best Δ
		Debate	MV-SFT	MV-GRPO	MV-KTO	MV-DPO	
Qwen2B	MATH	32.40	37.07 ± 3.07	39.00 ± 1.74	**46.47 ± 3.01**	42.60 ± 1.78	↑14.07
	GSM8K	49.60	50.53 ± 1.36	54.13 ± 2.02	**63.07 ± 0.64**	58.47 ± 1.62	↑13.47
	MathQA	24.20	26.27 ± 0.58	29.93 ± 4.99	**32.60 ± 0.72**	28.33 ± 0.31	↑9.13
Llama3B	MATH	37.80	35.33 ± 1.62	48.33 ± 2.19	**52.93 ± 0.99**	51.93 ± 1.67	↑15.27
	GSM8K	65.60	64.60 ± 1.59	68.60 ± 1.00	**73.13 ± 0.83**	71.67 ± 3.03	↑7.80
	MathQA	21.60	40.07 ± 2.00	48.73 ± 1.60	**64.00 ± 0.53**	63.13 ± 1.89	↑42.73
Phi4B	MATH	44.40	45.53 ± 2.53	**49.93 ± 1.33**	45.27 ± 0.70	46.73 ± 1.67	↑5.53
	GSM8K	79.60	78.93 ± 0.61	82.67 ± 1.81	82.47 ± 1.14	**84.73 ± 0.31**	↑5.13
	MathQA	49.60	50.87 ± 1.62	**63.07 ± 1.21**	51.53 ± 1.14	51.40 ± 1.44	↑13.47
Llama8B	MATH	32.80	34.13 ± 0.70	45.93 ± 1.03	53.93 ± 1.80	**59.67 ± 1.33**	↑26.87
	GSM8K	74.00	66.27 ± 1.01	81.53 ± 2.81	81.00 ± 1.97	**81.93 ± 1.51**	↑8.60
	MathQA	44.60	44.13 ± 1.10	57.27 ± 0.61	62.00 ± 2.03	**69.27 ± 1.55**	↑24.67



(a) Llama-3B on MATHQA **(b)** Llama-8B on MATH **(c)** Qwen-2B on GSM8K

Figure 2 Debate-aware RL improves all stages of multi-agent debate. Incorporating debate context in RL teaches agents to leverage prior arguments, improving final-round consensus. Stages: initial round average, initial round majority vote, final round average, final round majority vote.

(MATH (Hendrycks et al., 2021), GSM8K (Cobbe et al., 2021), MathQA (Amini et al., 2019), SVAMP (Patel et al., 2021), GPQA (Rein et al., 2023), and Commonsense-QA (Talmor et al., 2019)). We use 4-bit quantization with QLoRA (Dettmers et al., 2023), limit responses to 256 tokens, and instantiate multi-agent debate with $M = 3$ clones and $R = 2$ rounds—all design choices motivated by the computational constraints of multi-agent debate RL training on a single-node cluster. With temperature $\tau = 1.0$, this setup tests exploratory sampling under a budget adequate for efficient solvers on these benchmarks while remaining trainable with multiple concurrent agents. Detailed analysis of training efficiency and throughput is provided in App. Q. Critically, improvements persist when models are tested without token limits and without quantization (App. I.2, App. N), indicating that gains generalize beyond these specific settings. We compare (1) pre-trained models, (2) SFT baselines, and (3) MACA variants of GRPO, DPO, and KTO. We train and evaluate models on 1500/500 train/test splits for each dataset independently, unless otherwise specified, isolating task-specific performance improvements. We report mean agent accuracy with standard deviation across three seeds. Debate prompts, training parameters, multi-processing design, and other experimental details can be found in App. D.

5.1 Post-Training on MAD Improves Debate Performance

We evaluate MACA in the multi-agent debate setting, examining how post-training improves both the final consensus accuracy and individual agent performance across debate rounds.

Impact on multi-agent debate setting. We evaluate MACA in the multi-agent setting using final-round majority-vote accuracy (the consensus after debate). Baselines are (1) the base model's debate outcome (Du et al., 2023) and (2)

Table 2 Accuracy impact of MACA on single agent performance in zero-shot setting.

Model	Dataset	Base / SFT (Baseline)		RL (Our Methods)			Best Δ
		Base	MV-SFT	MV-GRPO	MV-KTO	MV-DPO	
Qwen2B	MATH	7.67	11.51 ± 0.60	18.09 ± 0.71	20.18 ± 0.67	$\mathbf{23.49 \pm 2.30}$	$\uparrow 15.82$
	GSM8K	23.00	24.84 ± 0.87	34.40 ± 2.08	$\mathbf{45.13 \pm 1.80}$	43.87 ± 1.92	$\uparrow 22.71$
	MathQA	5.00	5.25 ± 0.33	17.27 ± 2.88	$\mathbf{22.16 \pm 1.14}$	20.91 ± 0.47	$\uparrow 17.27$
Llama3B	MATH	27.87	25.89 ± 0.56	35.22 ± 0.44	40.64 ± 1.25	$\mathbf{40.71 \pm 0.08}$	$\uparrow 13.26$
	GSM8K	57.33	55.98 ± 0.68	52.40 ± 2.84	$\mathbf{65.76 \pm 1.44}$	64.98 ± 1.67	$\uparrow 8.80$
	MathQA	23.87	23.44 ± 0.73	30.09 ± 1.98	42.84 ± 0.67	$\mathbf{45.00 \pm 2.23}$	$\uparrow 21.13$
Phi4B	MATH	34.60	34.60 ± 0.82	$\mathbf{37.42 \pm 0.16}$	33.84 ± 0.78	34.62 ± 1.48	$\uparrow 2.82$
	GSM8K	67.27	69.58 ± 0.76	67.13 ± 3.60	75.60 ± 1.80	$\mathbf{76.87 \pm 0.36}$	$\uparrow 9.84$
	MathQA	34.87	34.04 ± 0.58	$\mathbf{45.52 \pm 2.19}$	33.91 ± 0.16	33.91 ± 0.50	$\uparrow 10.65$
Llama8B	MATH	22.93	23.16 ± 0.14	29.66 ± 1.27	39.42 ± 0.44	$\mathbf{46.00 \pm 0.35}$	$\uparrow 23.07$
	GSM8K	57.93	42.09 ± 1.28	62.45 ± 6.01	72.36 ± 1.34	$\mathbf{77.36 \pm 0.27}$	$\uparrow 19.43$
	MathQA	29.67	30.84 ± 0.60	33.07 ± 1.11	38.42 ± 1.22	$\mathbf{51.18 \pm 0.24}$	$\uparrow 21.51$

MV-SFT on majority traces (Subramaniam et al., 2025). As shown in Table 1, post-training on debate-derived signals improves ensemble accuracy across all models and datasets. Preference learning variants (MV-DPO and MV-KTO) provide the largest and most consistent gains, up to +42.73 percentage points, by directly optimizing the log-probability gap between trajectories, with MV-KTO performing best on smaller models ($\leq 3B$) and MV-DPO on larger ones ($4 - 8B$). Learning to differentiate between full reasoning trajectory pairs appears to better address credit assignment challenges in sparse final-answer supervision compared to both MV-GRPO's scalar rewards and MV-SFT's imitation learning, simultaneously improving answer accuracy while reinforcing higher-quality and more concise intermediate reasoning steps (App. L).

Impact on debate dynamics. Fig. 2 summarizes four metrics: initial round average single-agent accuracy, initial round ensemble majority vote (not used in the debate), final round average single-agent accuracy (conditioned on debate context/peer chains-of-thought), and final round ensemble majority vote (consensus). Post-training produces the largest gains where agents leverage peer feedback, since it directly teaches effective peer context utilization. We also observe that post-training mitigates debate-driven performance degradation observed in MATHQA for Llama-3B. The contrastive signal allows models to learn from consensus patterns even when exposed to flawed reasoning trajectories, correcting rather than amplifying poor grounding behaviors.

Impact on a single agent in zero-shot setting. Post-training on debate also improves the performance of the individual agents in the inital round (see Initial Average in Fig. 2): across 12 model-dataset pairs, MV-RL methods consistently outperform the Base and MV-SFT baselines (Table 2) in single-agent zero-shot (single trajectory) settings. Self-guided preference learning (MV-DPO and MV-KTO) outperforms scalar rewards via MV-GRPO for all models except Phi-4B. MV-DPO is best in 7/12 cases, while MV-KTO is better in some cases for smaller LMs.

5.2 Post-Training on MAD Improves Self-Consistency and Accuracy

We measure the effect of post-training on sampling consistency $s_t^{\theta,\tau}(x)$, the fraction of sampled trajectories that match the majority answer. As formalized in Sec. 3, we track the sampling consistency where $s_t^{\theta,\tau}(x)$ converges to the modal probability $S^{+}_{\theta,\tau}(x)$ as $t \to \infty$. For each model, we sample 20 trajectories on 500 held-out prompts and evaluate $s_t^{\theta,\tau}(x)$ for $t = 1$ to 20 (App. I.1). At $t = 1$, this metric primarily captures answer completeness, i.e., the percentage of responses that produce parseable answers within the token window, which post-training substantially improves. MV-DPO and MV-KTO achieve these gains through self-supervised preference learning alone, demonstrating that models can teach themselves more efficient reasoning without format rewards (App. L), which can otherwise be spurious (Huang et al., 2024; Srivastava et al., 2025). As sample size increases, the metric transitions to measuring true cross-sample agreement. The curves stabilize up to 27.6 percentage points above baseline, demonstrating that post-training increases answer concentration: models more consistently sample trajectories that converge despite high-temperature exploration. These improvements persist when tested without max token constraints (App. I.2).